National University of Computer and Emerging Sciences

Database Systems Lab (CL1002)

Date: 1st-December-2025

Course Instructor(s)

Ms. Sadaf Zehra

Lab Mid Exam (B)

Total Time: 120 minutes

Total Wtg: 50

Total Questions: 05

Semester: Fall-2025

Campus: Karachi

Department(s): CS

Submission Instructions:

- Return the question paper mention your **student ID on it.**
- Read each question completely before answering it. **There are 4 questions on 4 pages.**
- Attempt all the given questions. **All questions are carrying different points.**
- **Cheating in any case will lead to F-GRADE as per university rule.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.

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Student Name | Roll No | Section | Student Signature

Note: Submission will be (Query + output screenshot) for every question.

Question # 01: SQL Queries (LLO #: 1) ***[5 Weightage, Time: 15 Minutes]***

1. Write a query to find the following output:

	FIRST_NAME	DEPARTMENT_ID	SALARY
1	Neena	90	17000
2	Lex	90	17000
3	Bruce	60	6000
4	Daniel	100	9000
5	Alexander	30	3100
6	Matthew	50	8000
7	Karen	80	13500
8	Pat	20	6000
9	William	110	8300

2. Find the highest-paid employee in each department. Display the department name, salary, and employee first name.
3. Identify the departments that have more than 3 employees earning above their department's average.
4. Find employees who earn more than the average salary of all employees hired in the same year. Display the first name, hire year (asc), salary.
5. Write a query to find the departments where the difference between max and min salary exceeds 20% of the average salary. Display the department name, salary difference, and average salary.

Question # 02: Triggers & Transaction (LLO #: 2) **[20 Weightage, Time: 45 Minutes]**

a. TRIGGER

Courses Table

course_id	course_name	available_seats	total_seats
1	Data Science	5	30
2	Web Development	2	25
3	AI Fundamentals	8	40
4	Networking	1	20
5	Python Basics	10	35

Using the provided data of Coursess Table. Create a trigger named seat_threshold_check that fires BEFORE UPDATE on the Courses table. The trigger should monitor the available_seats column. If the new value (:NEW.available_seats) drops below 3, insert a record into the Course_Alert table with the following details:

- course_id
- current_seats
- alert_message = 'Low seat availability!'
- alert_date = SYSDATE

b. TRANSACTIONS

Create an Online Course Enrollment System. Firsts, create the following tables: students (student_id, name, email), courses (course_id, course_name, available_seats, total_seats), enrollments (enroll_id, student_id, course_id, enroll_date), payments (payment_id, enroll_id, amount), enrollment_log(log_id, enroll_id, action). Insert a new student into students. Check if the chosen course has available seats. If seats are available, create a new record in enrollments. Deduct one seat from the courses table. Insert a payment record in payments. Add a log entry into enrollment_log. Commit all changes if every step succeeds; otherwise, roll back the transaction.

Question # 03: PL/SQL (LLO #: 3) **[15 Weightage, Time: 40 Minutes]**

Books Table

book_id	book_title	author	total_copies	available_copies
1	Data Science 101	John Smith	10	5
2	Python Programming	Alice Brown	8	2
3	Database Systems	Mark Lee	12	7
4	Machine Learning	Sara Green	6	6

Borrow Table

borrow_id	book_id	borrow_date	quantity
1	1	2023-01-05	2
2	2	2023-01-12	1
3	3	2023-02-01	3

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1. A library system maintains two tables: Products (columns: book_id, book_title, author, total copies, and available_copies) and Borrow (columns: borrow_id, book_id, borrow_date, quantity). Create a stored procedure named RecordBorrow that takes two input parameters (p_book_id, p_quantity). The procedure should:

a. Check if the book exists in the Books table.
   - If not, display "Book not found."

b. If the book exists, check if the requested quantity exceeds the available copies.
   - If yes, display "Not enough copies available."

c. If enough copies are available:
   - Reduce the available copies in the Books table.
   - Insert a record into the Borrow table with current system date as borrow_date.

2. Create a stored function named GetTotalBorrowed that takes a book_id as input and returns the total borrowed quantity for that book (sum of all borrow quantities from the Borrow table).

Question # 04: MongoDB (LLO #: 4) ***[10 Weightage, Time: 20 Minutes]***

[{ "patient_id": "P001", "name": "Ayesha Khan", "age": 34, "gender": "Female", "disease": "Flu", "doctor": "Dr. Ali", "admitted": false, "bill": 1500 },

{ "patient_id": "P002", "name": "Hamza Ahmed", "age": 45, "gender": "Male", "disease": "Diabetes", "doctor": "Dr. Sara", "admitted": ~~true, "bill": 8000~~ },

{ "patient_id": "P003", "name": "Fatima Noor", "age": 29, "gender": "Female", "disease": "Fracture", "doctor": "Dr. Ahmed", "admitted": true, "bill": 12000 },

{ "patient_id": "P004", "name": "Omar Zain", "age": 51, "gender": "Male", "disease": "Hypertension", "doctor": "Dr. Sara", "admitted": true, "bill": 9500 },

{ "patient_id": "P005", "name": "Nida Raza", "age": 38, "gender": "Female", "disease": "Flu", "doctor": "Dr. Ali", "admitted": false, "bill": 1200 },

{ "patient_id": "P006", "name": "Bilal Aslam", "age": 60, "gender": "Male", "disease": "Heart Disease", "doctor": "Dr. Ahmed", "admitted": true, "bill": 25000 },

{ "patient_id": "P007", "name": "Sana Qureshi", "age": 27, "gender": "Female", "disease": "Migraine", "doctor": "Dr. Fatima", "admitted": false, "bill": 1800 },

{ "patient_id": "P008", "name": "Kashan Malik", "age": 33, "gender": "Male", "disease": "Flu", "doctor": "Dr. Ali", "admitted": true, "bill": 3000 },

{ "patient_id": "P009", "name": "Zoya Tariq", "age": 41, "gender": "Female", "disease": "Diabetes", "doctor": "Dr. Sara", "admitted": true, "bill": 10500 },

{ "patient_id": "P010", "name": "Usman Farid", "age": 50, "gender": "Male", "disease": "Hypertension", "doctor": "Dr. Sara", "admitted": false, "bill": 6000 }]

1. Update the bill of all patients admitted with "Flu" by 20%, and set "admitted": false after update.
2. Delete all records of patients who are not admitted and whose bill is less than 2000.
3. Find all patients treated by "Dr. Sara" with a bill greater than 7000, sorted by bill descending.
4. Update the bill of patients over age 50 by 10% if they are currently admitted.

5. Find all patients who are admitted and their disease is either "Heart Disease" or "Fracture".
6. Delete all patients whose disease is "Flu" and admitted status is false.
7. Update "Dr. Ahmed's" patients' bill by reducing 15%, but only if the bill is greater than 10000.
8. Find all patients where gender is "Female" and bill is between 1000 and 10000.
9. Update the name "Hamza Ahmed" → "Hamza A. Khan" and increase his bill by 25%.
10. Delete all records of patients older than 55 years and not admitted.

BEST OF LUCK☺